Design, Analysis, and Drive Control of a Nissan Leaf Referenced IPM Motor Using FluxMotor and PSIM

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Abstract

In this study, an interior permanent magnet (IPM) synchronous traction motor, using the Nissan Leaf as a reference, was modeled in the FluxMotor environment, and its fundamental electromagnetic and thermal analyses were performed. Open-circuit analyses, d-q axis parameter maps, and operating point evaluations were conducted for the motor. The obtained results demonstrate that the design exhibits an IPM motor characteristic and provides a suitable structure for d-q axis-based control approaches. However, the torque ripple, iron losses in the high-speed region, and thermal results have revealed aspects of the design that are open to improvement. In the second part of the study, a drive model was established in the PSIM environment for the same motor type; speed control, d-q current control, maximum torque per ampere (MTPA), field weakening, and maximum torque per voltage (MTPV) approaches were investigated. Simulation results indicate that the established drive structure is capable of tracking the reference speed and exhibits a control behavior consistent with the motor characteristics.

Keywords: IPM motor, Nissan Leaf, FluxMotor, PSIM, MTPA, field weakening, back-EMF, thermal analysis

1. Objective of the Project

In this study, using the Nissan Leaf motor as a reference, it is aimed to model a traction motor with an interior permanent magnet (IPM) synchronous motor structure, to perform its fundamental electromagnetic and thermal analyses, and to investigate a suitable drive-control structure for this motor type. The study consists of two main sections; in the first section, the motor topology was created in the FluxMotor environment; the geometry, winding structure, magnet placement, material selection, and cooling assumptions were defined. Subsequently, the open-circuit behavior, d-q axis parameter maps, operating point results, and steady-state thermal behavior were evaluated. In the second section, a drive model was established in the PSIM environment for the same motor type, and control structures based on maximum torque per ampere (MTPA), field weakening, and maximum torque per voltage (MTPV) were investigated.

The main objective of the study is not only to geometrically reconstruct an IPM motor similar to the Nissan Leaf, but also to demonstrate the relationship between the electromagnetic characteristics of this structure and the drive-control approach. In this context, cogging torque, back-EMF, flux linkage, inductance maps, fundamental operating point performance, and thermal constraints were evaluated together. Based on the obtained results, it was aimed to determine the strengths and weaknesses of the selected design and to interpret them particularly in terms of high-speed performance, torque ripple, and thermal behavior.

This report aims to present together the fundamental engineering decisions that may be encountered in the initial design iteration of a traction motor and the consequences of these decisions. Therefore, rather than presenting a perfect final design; the study is approached as an applied study that systematically examines the design, analysis, and control processes of a reference IPM motor topology.

2. Reason for Selecting the Nissan Leaf-Referenced IPM Motor

The main reason for selecting the interior permanent magnet (IPM) synchronous motor structure of the Nissan Leaf as the reference motor in this study is that this topology offers a solution widely used in electric vehicle traction applications and has a well-established equivalent in the literature. The primary characteristics expected from traction motors used in electric vehicles are high torque density, the ability to operate over a wide speed range, high efficiency, and suitable controllability. IPM motors hold a significant place in automotive applications because they can fulfill these requirements.

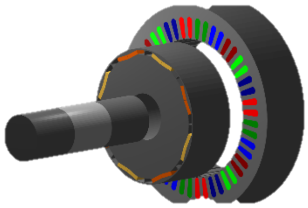

Figure 1. Radial View of the Motor

Compared to surface-mounted permanent magnet synchronous motors, embedding the magnets inside the rotor in the IPM structure allows for the formation of rotor saliency. In this way, not only magnet-derived torque but also a reluctance torque contribution can be obtained. This provides a significant advantage, particularly in terms of high efficiency and producing higher torque with lower current. Furthermore, the interior permanent magnet structure presents a characteristic that is more suitable for field weakening control in high-speed regions, thereby establishing an operational range favorable for electric vehicle driving conditions.

Another reason for referencing the Nissan Leaf motor is that it is a well-known electric vehicle motor in the industry and possesses the typical characteristics of the IPM topology. The objective within the scope of this study is not to reproduce the Leaf motor exactly at a commercial level, but rather to create an IPM motor structure with similar topological features and to perform its fundamental analyses. This approach enables both an understanding of the electromagnetic design process and the investigation of the drive-control structure on an appropriate motor model.

In the selected motor structure, an inner rotor, permanent magnets, a hairpin winding, and a three-phase supply arrangement are considered together. This preference is consistent with the structural trends commonly observed in modern traction motors. However, it has been acknowledged from the outset that the design created in this study is an initial iteration and is open to improvement, particularly regarding aspects such as cooling, harmonic content, and torque ripple. Therefore, the motor selection is considered a reference platform suitable for analysis and evaluation, rather than a solution optimized at the final product level.

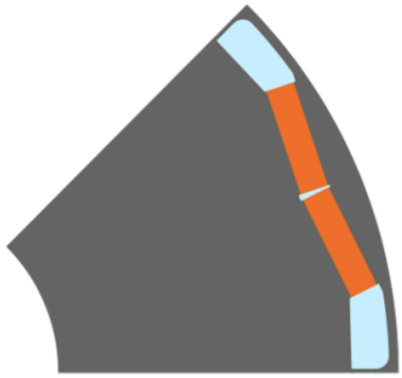

Figure 2. Rotor magnet view

3. Motor Design and Main Technical Specifications

In this section, the fundamental geometric, electromagnetic, and structural characteristics of the generated Nissan Leaf-referenced IPM motor design are summarized. The motor model was created in the FluxMotor environment using a three-phase, interior permanent magnet synchronous machine topology. During the design process, the stator and rotor dimensions, magnet placement, winding structure, materials, and basic cooling assumptions were considered together.

When the structural data of the motor are examined, it is seen that the stator outer diameter is 198 mm, the stator inner diameter is 132 mm, and the active length is 50 mm. The rotor outer diameter is defined as 130 mm, and the rotor inner diameter is 44.45 mm. The air gap length was selected as 1 mm, and the motor was modeled with 48 slots and 8 poles. This selection provides a suitable slot-pole combination for three-phase IPM traction motors and establishes a structure appropriate for the d-q axis-based control approach.

On the rotor side, an imi_VBlock_01E type V-block magnet arrangement was used. REF.NdFeB_1370_1273 was selected as the magnet material. The remanent flux density of this material at a reference temperature of 20°C is 1.37 T, and its intrinsic coercive field is 1.273×10^6 A/m. The selected interior permanent magnet structure allows for the formation of rotor saliency and, consequently, the emergence of a reluctance torque contribution. The V-shaped arrangement of the magnets within the rotor is one of the fundamental design decisions that directly affect the air gap flux distribution and high-speed behavior.

On the stator side, the os_PllTooth_05E slot structure and hairpin winding technology were preferred. The winding connection is defined as star (wye), and the system has a three-phase structure. Two parallel paths were used for each phase. A rectangular cross-section conductor was preferred in the winding; the conductor width is defined as 1.2 mm, the height as 2.5 mm, and the half-space between conductors as 0.1 mm. In the winding layout, the number of turns per coil is 1, and the total number of series turns per phase is 24. Although this structure is suitable for the hairpin approach used in traction motors, it should be additionally evaluated in terms of the in-slot conduction area and the free space ratio.

When the winding results are examined, it is seen that the total slot area is 109.31 mm², and only an 18 mm² portion of this corresponds to the conductive region. Accordingly, the gross fill factor was obtained as 16.47%, and the net fill factor as 22.84%. While the high amount of free space within the slot does not pose a problem in terms of electrical layout, it can create a weak structure in terms of thermal conduction. In particular, the absence of impregnation and the low equivalent radial thermal conductivity within the slot have a determining effect on the thermal behavior of this design.

The fundamental harmonic winding factor of the winding was obtained as 0.9659. This high value indicates that the winding layout is appropriate in terms of the fundamental component. The distribution factor and pitch factor are also close to 1, yielding a positive result for the production of the fundamental electromagnetic component. However, the fact that no skew was applied on the rotor and stator sides requires additional evaluation regarding harmonic content and torque ripple.

In the mechanical structure of the motor, the shaft and frame components are defined with steel-based materials. The total motor mass is calculated to be approximately 22.308 kg. Of this, 16.093 kg belongs to the stator and 6.215 kg to the rotor section. The rotor moment of inertia was found to be $9.634x10^{-3} kg{\cdot}m^2$. These data are fundamental quantities that can be used later in terms of mechanical load and dynamic behavior when establishing the drive model.

In the design, M330_35A electrical steel material was used for the stator and rotor magnetic circuits. This material has a non-linear B-H characteristic and is defined by a 0.35 mm lamination thickness and a 95% stacking factor. Copper was selected as the conductive material, and Nomex 130 as the insulation material. This material selection allows for the establishment of a consistent structure between the electromagnetic analysis and the thermal model.

In this initial design iteration, the geometric and electromagnetic structure of the motor was created to represent a Nissan Leaf-type IPM motor, but the cooling structure was kept simple. There are no cooling fins on the frame, no active cooling circuit has been defined, and heat transfer to the external environment is limited to natural convection and radiation. Therefore, although the motor presents an electromagnetically workable topology, its thermal performance must be evaluated separately.

Table 1. Main geometric specifications of the motor.

Parameter	Symbol / Description	Value
Stator outer diameter	$\boldsymbol{D_{so}}$	198 mm
Stator inner diameter	$\boldsymbol{D_{si}}$	132 mm
Rotor outer diameter	D_{ro}	130 mm
Rotor inner diameter	D_{ri}	44.45 mm
Active length	$\boldsymbol{L}$	50 mm
Air gap length	$\boldsymbol{g}$	1.0 mm
Number of slots	$\boldsymbol{N_s}$	48
Number of poles	$\boldsymbol{2p}$	8
Number of phases		3
Topology		IPM

Table 2. Winding, magnet, and material properties

Parameter	Description	Value
Winding technology		Hairpin
Winding connection		Wye
Number of parallel paths		2
Number of turns per coil		1
Number of series turns per phase		24
Number of conductors per phase		96
Conductor type		Rectangular
Conductor width		1.2 mm
Conductor height		2.5 mm
Half-space between conductors		0.1 mm
Gross fill factor	Gross fill factor	%16.47
Net fill factor	Net fill factor	%22,84
Fundamental harmonic winding factor	$\boldsymbol{k_w}$	0.9659
Fundamental harmonic distribution factor	$\boldsymbol{k_d}$	0.9694
Fundamental harmonic pitch factor	$\boldsymbol{k_p}$	0.9964
Skew factor	$\boldsymbol{k_{skew}}$	1.0
Mıknatıs Magnet material		NdFeB_1370_1273
Remanent flux density	$\boldsymbol{B_r}$	1.37 T
İntrinsik Intrinsic coercive fieldalan	$\boldsymbol{H_{cj}}$	$\mathbf{1.273 \times 10^6}$ A/m
Stator / rotor electrical steel material		M330_35A
Conductor material		Copper
Insulation material		Nomex 130

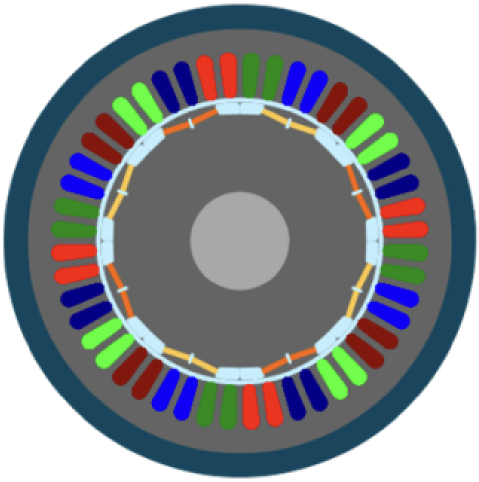

Figure 3. Radial View of the Motor

Figure 4. Slot Geometry

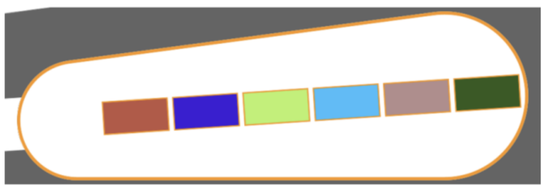

Figure 5. Slot Fil

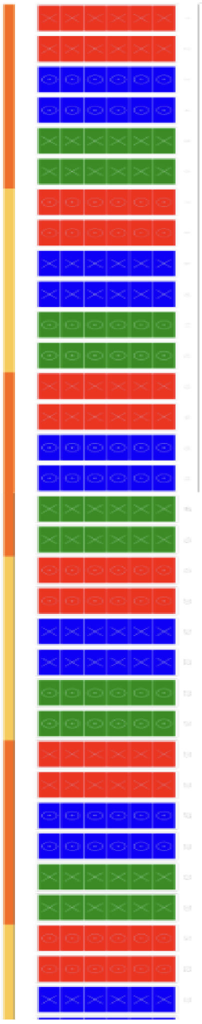

Figure 6. Hairpin Winding Connections

4. Cooling Structure and Thermal Assumptions

The thermal behavior of the motor is directly dependent not only on the electromagnetic loss values but also on the selected frame structure, cooling approach, and internal heat conduction paths. Therefore, for a sound interpretation of the thermal analysis results, the cooling assumptions used in the model must be specified separately. The motor model created in this study was evaluated with a simple and passive cooling structure.

A circular frame structure was used in the frame design. The frame thickness was selected as 10 mm, and a 50 mm extension was defined on the connection and opposite connection sides. However, no cooling fins were defined on the frame. Similarly, an active cooling circuit or liquid cooling channel was not included in the model. For this reason, the heat transfer of the motor to the external environment remains limited to natural convection and radiation occurring only over the frame surfaces.

When the external cooling model is examined, it is seen that the external fluid is air and the convection mode is defined as natural convection. The external ambient temperature was set to 20 °C, and the radiation emissivity of the frame surface to the environment was selected as 0.8. These assumptions mean a lower heat dissipation capacity compared to fan-assisted or liquid-jacket cooling. Consequently, this design presents a thermally limiting structure at operating points with high losses.

Similarly, a natural convection approach was used on the internal cooling side. The internal fluid was defined as air; convection and radiation effects in the air gap, rotor end regions, and end-space regions were included in the model. However, there is no forced air flow or active internal cooling mechanism in these regions either. It is evaluated that this assumption has a significant limiting effect, particularly on the transfer of the heat generated in the end winding regions to the external environment.

The slot thermal model is one of the critical points of this design. Impregnation was not applied, and potting was not defined in the winding structure. An automatic equivalent conductivity model was used for the copper, insulation, and air regions within the slot. According to the values given in the report, the equivalent radial and orthoradial thermal conductivity of the slot region is at a rather low level. In contrast, the axial conductivity was calculated as 64.549 W/K/m. This situation indicates that the heat generated in the winding can be transferred more easily in the axial direction; however, it cannot be effectively transferred radially from inside the slot to the stator tooth and the frame.

The low gross fill factor (16.47%) and net fill factor (22.84%) in the winding structure are also noteworthy in terms of thermal behavior. Although having a high amount of free space within the slot seems advantageous for cooling at first glance, the fact that this gap is not filled with an effective thermal filler material and largely behaves like air does not improve heat transfer in reality. On the contrary, it contributes to an increase in the winding temperature by lowering the equivalent in-slot thermal conductivity.

The contact resistances defined in the model also affect the internal heat transfer paths. The magnetic circuit-frame, magnetic circuit-slot, magnetic circuit-magnet, and magnetic circuit-shaft interfaces are defined with specific thickness values. The additional thermal resistances created by these interfaces are particularly important regarding the rotor magnet temperature and the heat transfer from the stator copper toward the frame. In addition, specific conduction resistances are present in the bearing and end-cap connections.

Consequently, the thermal model used in this study represents a passively cooled and structurally simple motor frame. Since improving elements such as cooling fins, active liquid cooling, impregnation, and potting were not included in the model, the thermal analysis results must be evaluated under the natural and limited cooling capacity of the selected structure. Therefore, the high temperature values obtained in the subsequent sections should be interpreted not only in relation to the loss levels but also in the context of the weak cooling infrastructure defined here.

Table 3. Cooling structure and thermal model assumptions

Parameter	Description	Value
Frame topology		Circular
Frame thickness		10 mm
Connection side extension	C.S. extension	50 mm
Opposite connection side extension	O.C.S. extension	50 mm
Cooling fin	Fin topology	None
Active cooling circuit	Cooling circuit	None
External cooling mode		Natural
External fluid		Air
External ambient temperature	External fluid temperature	20 °C
External radiation mode	Radiation mode	Automatic
Frame emissivity	Frame to infinite emissivit	0.8
Internal cooling mode	Internal convection mode	Natural
Internal fluid		Air
Airgap thermal model	Airgap convection/radiation	Automatic
End-space model	Connection side / opposite side	Automatic
Mechanical loss model	In thermal test	0 W
Magnet losses	In thermal test	0 W
Rotor iron losses	In thermal test	0 W

Table 4. Slot thermal model and winding-related thermal parameters

Parameter	Description	Value
Impregnation	Coil impregnation	No
Potting	C.S. / O.C.S. potting	None
Gross fill factor		%16.47
Net fill factor		%22,84
Slot area		109.31 mm²
Conductive area		18.0 mm²
Free area		84.345 mm²
Equivalent radial/orthoradial thermal conductivity	Radial and orthoradial equivalent conductivity	4.916×10^{-2} W/K/m

Parameter	Description	Value
Equivalent axial thermal conductivity	Axial equivalent conductivity	64.549 W/K/m
Electrical conductor thermal conductivity	Copper thermal conductivity	385 W/K/m
Insulation thermal conductivity	Nomex thermal conductivity	0.21 W/K/m
Internal fluid thermal conductivity	Air thermal conductivity	2.568×10^{-2} W/K/m
Slot – C.S. end winding thermal resistance	Slot-C.S. end winding resistance	1.096×10^{-1} K/W
Slot – O.C.S. end winding thermal resistance	Slot-O.C.S. end winding resistance	1.127×10^{-1} K/W

As seen in Table 3 and Table 4, the motor model was investigated with a passively cooled structure. There are no cooling fins or active cooling circuit on the frame, and internal and external heat transfer is limited to natural convection and radiation. Furthermore, the absence of impregnation and the low equivalent radial thermal conductivity in the slot region make it difficult for the heat generated in the winding to be effectively transferred to the frame. Therefore, the obtained thermal results are affected by the cooling assumptions as much as by the selected electromagnetic loss levels.

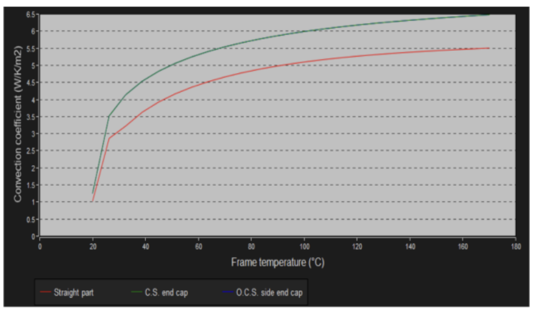

Figure 7. Frame natural convection coefficient / resistance graph

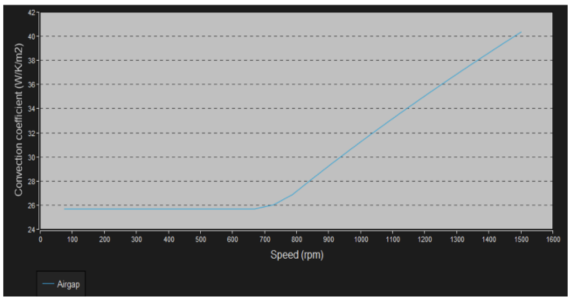

Figure 8. Airgap convection coefficient versus speed

5. Open Circuit Electromagnetic Analyses

As the first evaluation step of the motor design, open-circuit electromagnetic analyses were performed. Open-circuit tests were used to investigate the electromagnetic behavior generated solely by the rotor magnets and the stator slot structure without applying phase current. In this context, cogging torque, back-EMF, and air gap flux density were evaluated together. Since open-circuit analyses reveal the fundamental magnetic characteristics of the motor without the presence of load effects, they hold a significant place in the initial verification stage of the design.

5.1 Cogging Torque Analysis

Cogging torque analysis was conducted to investigate the torque ripple that occurs depending on the relative positions of the rotor magnets and stator slots. In this study, the analysis was performed at a magnet temperature of 20 °C, and the torque variation depending on the rotor angle was calculated over one cogging period. According to the obtained results, the peak-to-peak value of the cogging torque was found to be 3.657 N·m. When the graph dependent on the rotor position is examined, it is seen that the waveform is largely determined by low-order harmonics.

Examining the cogging torque harmonic analysis, it is observed that the 1st harmonic component is dominant at 1.634 N·m, the 2nd harmonic component remains at the 0.469 N·m level, and higher-order harmonics decrease significantly. This situation indicates that the waveform has a characteristic defined by low-order components rather than a completely irregular structure. However, the absolute cogging torque level needs to be evaluated in terms of vibration, acoustic behavior, and torque smoothness in the low-speed and starting regions.

When the open-circuit flux density distribution is examined, flux density values approaching 2 T were obtained, especially in the rotor bridge and rotor pole shoe regions. It is understood that a local saturation effect emerges due to the presence of narrow magnetic cross-sections in these regions. The fact that the internal rotor geometry and the absence of skew, as much as the magnet structure, seem to be effective in the cogging torque exceeding a certain level.

5.2 Back-EMF Analysis

The back-EMF analysis was performed at a mechanical speed of 750 rpm and a magnet temperature of 20 °C. The winding connection was defined as star (wye). According to the obtained results, the back-EMF constant was found to be $k_E = 0.2396\ \mathrm{V \cdot s/rad}$. The 1st harmonic peak value of the phase voltage is 10.229 V, and its RMS value is 7.233 V. For the line-to-line voltage, these values were calculated as 17.74 V and 12.544 V, respectively.

The total harmonic distortion of the phase voltage was obtained as approximately 10.34%, while the total harmonic distortion of the line-to-line voltage was approximately 5.22%. This situation indicates that the line-to-line voltage is smoother and more sinusoidal compared to the phase voltage because the triplen harmonics are largely suppressed in the line-to-line voltage. When the harmonic content is examined, although the fundamental component is dominant, it is observed that the 3rd, 5th, and 9th harmonics are particularly evident in the phase voltage.

The open-circuit air gap flux density distribution deviates from the ideal sinusoidal structure and mostly exhibits a waveform close to a flat-topped, trapezoidal character. The components of the air gap flux density other than the fundamental harmonic are consistent with the observed back-EMF harmonics. This situation can be explained by the magnet geometry, slot structure, and local flux concentration in the rotor pole shoe regions. In the phase flux linkage results, it is seen that the d-axis flux is dominant, while the q-axis flux is at a negligible level under open-circuit conditions. This indicates that the magnet flux is formed in alignment with the rotor magnetic axis.

5.3 Evaluation of Air Gap Flux Density and Magnetic Structure

Under open-circuit conditions, the average flux density in the air gap was found to be approximately 5.345×10^{-1} T, and the peak value was approximately 0.929 T. While the flux density in the stator tooth and stator yoke regions reaches levels of approximately 1.64–1.67 T, values exceeding the 2 T level were obtained in the rotor bridge and rotor pole shoe regions. This result indicates that there is a tendency for local saturation, especially in the narrow magnetic cross-sections of the rotor.

Although this local saturation effect is not considered a serious design flaw on its own, it should be viewed as a significant factor affecting the back-EMF waveform, harmonic content, and cogging behavior. In particular, the absence of skew indicates that no additional precautions have been taken to reduce these harmonic effects and torque ripples.

Overall, the open-circuit electromagnetic analysis results reveal that the selected IPM structure exhibits physically consistent magnetic behavior. The phases show a balanced structure, the fundamental harmonic component emerges as dominant, and the air gap flux is generated at the expected magnitudes. However, it is understood that the structure is open to improvement in terms of cogging torque and harmonic content.

Table 5. Open-circuit cogging torque results

Parameter	Value
Analysis period	7.5 deg
Cogging torque, peak-to-peak	3.657 N·m
1st harmonic	1.634 N·m
2nd harmonic	0.469 N·m
3rd harmonic	0.0416 N·m

Table 6. Open-circuit back-EMF results (750 rpm)

Parameter	Value
Speed	750 rpm
k_E	0.2396 V·s/rad

Parameter	Value
Phase voltage, 1st harmonic peak	10.229 V
Phase voltage, 1st harmonic RMS	7.233 V
Phase voltage THD	%10.34
Line-to-line voltage, 1st harmonic peak	17.74 V
Line-to-line voltage, 1st harmonic RMS	12.544 V
Line-to-line voltage THD	%5.22

Table 7. Open-circuit air gap and iron flux density results

Parameter	Value
Air gap flux density, ARV	0.5345 T
Air gap flux density, 1st harmonic RMS	0.6625 T
Air gap flux density, peak	0.9289 T
Stator tooth, max	1.665 T
Stator foot tooth, max	1.661 T
Stator yoke, max	1.639 T
Rotor yoke, max	1.387 T
Rotor web, max	0.4426 T
Rotor bridge, max	2.023 T
Rotor pole shoe, max	2.141 T

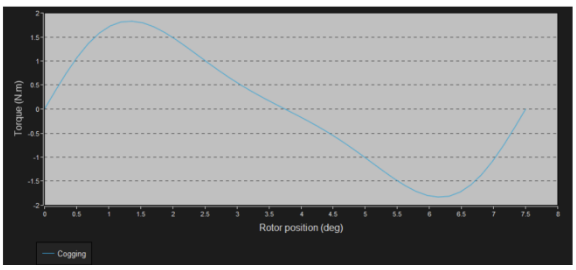

Figure 9. Variation of cogging torque with rotor angle

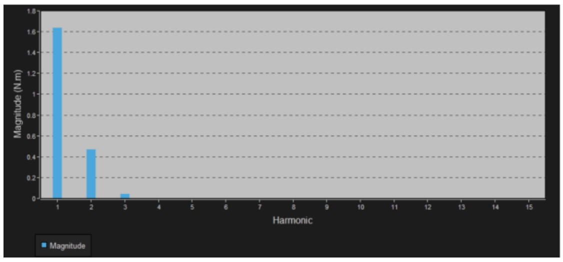

Figure 10. Cogging torque harmonic analysis

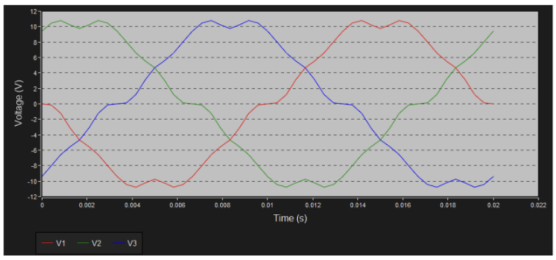

Figure 11. Variation of phase voltage with time (open circuit)

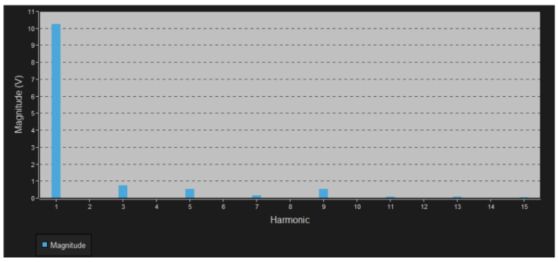

Figure 12. Phase voltage harmonic analysis (open circuit)

Harmonic	Magnitude (V)	Phase (deg)	Frequency (Hz)
1	10.229	89.999	50.0
3	7.452 E-1	-89.993	150.0
5	5.086 E-1	-90.018	250.0
7	1.292 E-1	90.098	350.0
9	5.243 E-1	-90.247	450.0
11	6.744 E-2	-90.113	550.0
13	7.974 E-2	-89.366	650.0
15	3.305 E-2	88.957	750.0

Figure 13. Line-to-line voltage harmonic analysis (open circuit)

Harmonic	Magnitude (V)	Phase (deg)	Frequency (Hz)
1	17.74	119.99	50.0
3	3.168 E-2	67.5	150.0
5	8.936 E-1	-121.081	250.0
7	1.425 E-1	121.272	350.0
9	1.116 E-2	-161.888	450.0
11	1.273 E-1	-129.626	550.0
13	1.471 E-1	-62.787	650.0
15	2.689 E-2	-101.709	750.0

Figure 14. Line-to-line voltage harmonic analysis (open circuit)

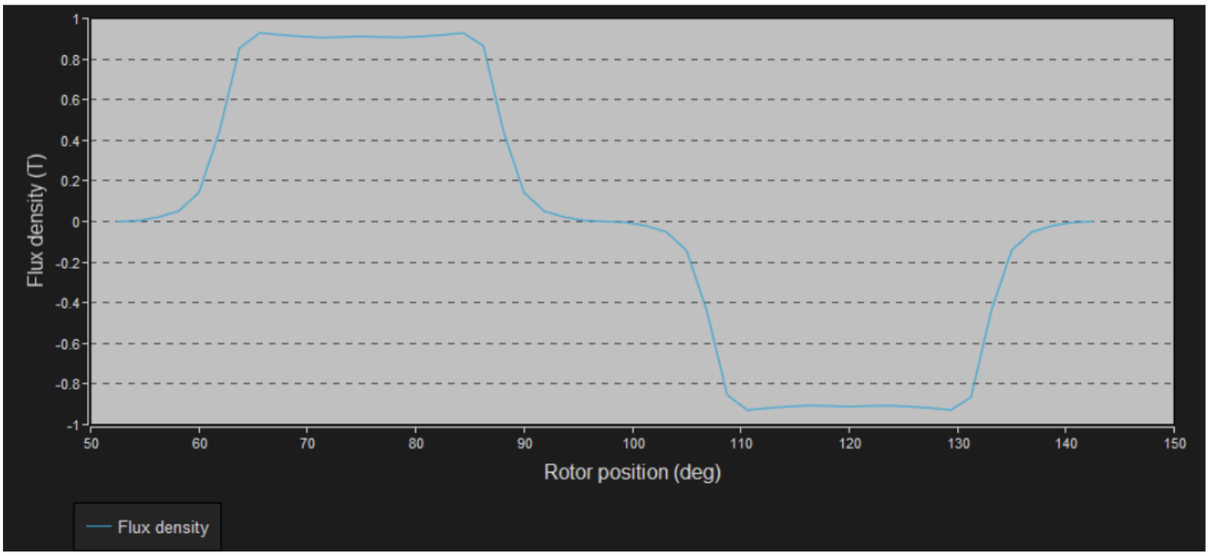

Figure 15. Flux density in the airgap versus rotor angular position – open circuit

The open-circuit results verify the fundamental magnetic behavior of the motor, but indicate that the design is open to improvement in terms of harmonic content and cogging torque. Therefore, in the next stage, the behavior under load needs to be evaluated in more detail by examining the d-q axis parameter maps of the motor.

6. d-q Axis Model and Parameter Maps

To investigate the controllability and electromagnetic behavior of IPM motors in different operating regions, d-q axis-based parameter maps are an important tool. In this study, the flux linkage, dynamic and static inductances, electromagnetic torque production, and loss distribution of the motor in the $J_d - J_q$ plane were extracted in the FluxMotor environment. Thus, not only the open-circuit characteristics of the motor but also the non-linear magnetic behavior it exhibits under current were evaluated.

While generating the maps, the motor was examined in the second quadrant, that is, on the negative J_d and positive J_q axes. This selection represents the motoring operating region of the IPM motor and specifically the area where the MTPA characteristic is observed. The current definition mode was selected as density, the maximum line current was defined as 36 A RMS, and the maximum current density as 6 A/mm². All parameter maps were obtained under a constant temperature condition of 20 °C.

6.1 Flux Linkage Maps

When the d-axis flux linkage (Φ_d) maps are examined, it is seen that the flux linkage varies largely depending on the J_d current. It is understood that as J_d goes to more negative values, the d-axis flux linkage decreases, and as J_d approaches zero, it increases. Conversely, the effect of the J_q current on Φ_d is more limited. This result indicates that the d-axis flux is primarily determined by the magnet flux and the d-axis current.

The q-axis flux linkage (Φ_q) maps, on the other hand, vary predominantly along the J_q axis. It is observed that as J_q increases, the Φ_q value rises significantly. Although the effect of the J_d variation on Φ_q is weaker, it is not completely negligible. This situation, while confirming that the q-axis is the main torque-producing axis, also demonstrates that magnetic interaction between the axes is present at high current levels.

When the flux linkage maps are evaluated together, it is seen that there is a distinct physical separation between the d and q axes of the motor. However, some curve deviations and small changes occurring as the current increases indicate that the motor has a completely non-linear magnetic circuit. Especially at high current levels, this situation points to the involvement of cross-saturation effects.

6.2 Dynamic and Static Inductance Maps

Dynamic inductance maps are important in terms of showing the magnetic response given by the motor to small-signal current variations. In the obtained results, it is seen that the d-axis dynamic inductance (L_d) value changes depending on the J_d and J_q currents. This variation indicates the presence of a saturation effect in the magnetic circuit. While the inductance is expected to remain constant in a fully linear machine, observing a significant variation dependent on the current level in this motor is a clear indicator of non-linear behavior.

Similarly, the q-axis dynamic inductance (L_q) is not constant but varies depending on current levels. When the maps are examined, it is generally seen that the L_q value is greater than the L_d value. This result shows the presence of rotor saliency specific to the interior permanent magnet structure. In IPM motors, the $L_q > L_d$ relationship is one of the fundamental conditions for the generation of reluctance torque contribution.

When evaluated together with the datasheet results, it is seen that in the unsaturated condition, the L_d and L_q values are quite close to each other, but under the operating point, this difference becomes more pronounced due to the saturation effect. This suggests that rotor saliency is strengthened not only by the geometry but also by the magnetic saturation effect. In other words, the IPM characteristics of the motor are influenced not only by the magnet placement but also by the magnetic field distribution formed under current.

Static inductance maps also exhibit a similar trend. Static inductances can be considered as the total flux linkage divided by the current, and the maps show that the inductance variation is more pronounced in certain operating regions of the motor. The combined evaluation of dynamic and static inductances indicates that the fixed-parameter linear model approach in control design may remain limited.

6.3 Cross-Inductances and Cross-Coupling Interaction

The cross-inductance components L_{dq} and L_{qd} given in the maps indicate the coupling between the d and q axes. In an ideal linear model, these components are expected to be very close to zero. In this study, it is observed that the cross-inductances are at quite low levels, but they are not completely zero. This result demonstrates that there is a certain magnetic interaction between the axes, but this effect remains more limited compared to the main inductance components.

When considered together with the small variations observed specifically in the $\Phi_d - J_q$ curve, it is understood that the q-axis current has a small but measurable effect on the d-axis flux linkage. Such deviations indicate that cross-saturation effects at high current levels should be taken into account regarding control performance.

6.4 Electromagnetic Torque and Loss Maps

The electromagnetic torque map shows that torque increases largely with the J_q current. As J_q increases, the torque contour lines rise regularly, while the variation in the J_d axis affects this increase secondarily. This result confirms that the q-axis current is the primary torque production component. However, it is observed that negative J_d values support torque in certain regions; this situation demonstrates the presence of a reluctance torque contribution in the IPM structure.

When the stator iron loss and total loss maps are examined, it is seen that losses increase particularly in the high J_q and lower negative J_d regions. The stator Joule loss map, as expected, rises with the increase in current magnitude. These results indicate that when selecting the operating point, not only maximum torque production but also loss distribution must be taken into consideration.

One of the provided curves, the $\Phi_d - J_q$ relationship, also indicates that the d-axis flux linkage is only affected to a limited extent by the q-axis current. The fact that the curve is largely close to horizontal demonstrates that the cross-coupling between the axes is weak, but the fact that it is not perfectly horizontal indicates that non-linear interactions cannot be completely ignored.

6.5 General Evaluation

The d-q axis-based parameter maps clearly demonstrate that the created motor model possesses an IPM motor characteristic. In particular, the $L_q > L_d$ relationship confirms the presence of rotor saliency and the ability to generate a reluctance torque contribution. Furthermore, the variation of inductances depending on the current and the fact that cross-inductances are not exactly zero reveal that the motor exhibits non-linear magnetic behavior.

These results indicate that in the control design of the motor, a map-based or operating region-sensitive approach would be more accurate than a simple model with constant parameters. Indeed, the preference for MTPA, field weakening, and MTPV strategies in the drive model established in the PSIM environment in the next stage is a choice consistent with this electromagnetic character.

Table 8. Key observations regarding the d-q axis model and map analysis

Investigated Quantity	Observation	Comment
Φ_d map	Primarily varies with I_d	d-axis flux is dominantly dependent on d-axis current
Φ_q *map*	Primarily varies with J_q	q-axis flux is consistent with the torque-producing axis
L_d *map*	Varies depending on current	Saturation effect is present in the d-axis
L_q *map*	Varies depending on current, generally $L_q > L_d$	IPM saliency and reluctance torque contribution
L_{dq},L_{qd} *maps*	Small but not zero	Cross-saturation / cross-coupling interaction is present
Torque map	Torque increases largely with J_q	q-axis current is the primary torque component
Loss maps	Total loss increases as current increase	Osses must be considered in operating point selection

Table 9. d-q model parameters obtained from datasheet analysis (base speed point)

Parameter	Value
D-axis inductance, $\boldsymbol{L_d}$	$\mathbf{7.984} \times \mathbf{10^{-5}}$ H
Q-axis inductance, $\boldsymbol{L_q}$	$\mathbf{1.084} \times \mathbf{10^{-4}}$ H
D-axis flux	$\mathbf{3.21} \times \mathbf{10^{-2}}$ Wb
Q-axis flux	$\mathbf{6.761} \times \mathbf{10^{-3}}$ Wb
D-axis phase current, peak	-4.61 A
Q-axis phase current, peak	50.703 A
D-axis phase voltage, peak	-71.252 V
Q-axis phase voltage, peak	310.36 V

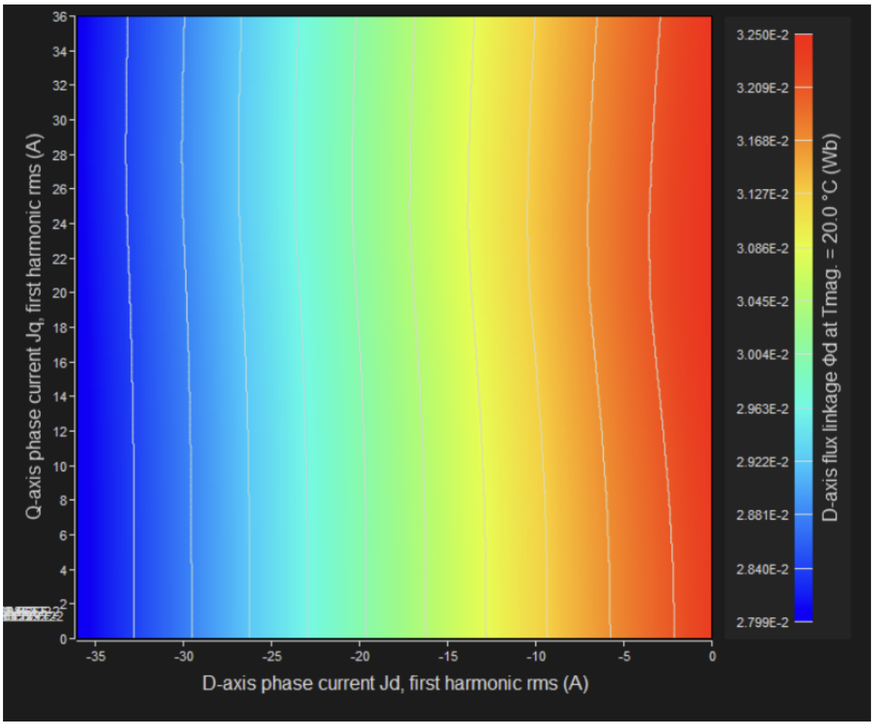

Figure 16. D-axis flux linkage $\boldsymbol{\Phi}_{\boldsymbol{d}}$ in $\boldsymbol{J}_{\boldsymbol{d}}$-$\boldsymbol{J}_{\boldsymbol{q}}$ area

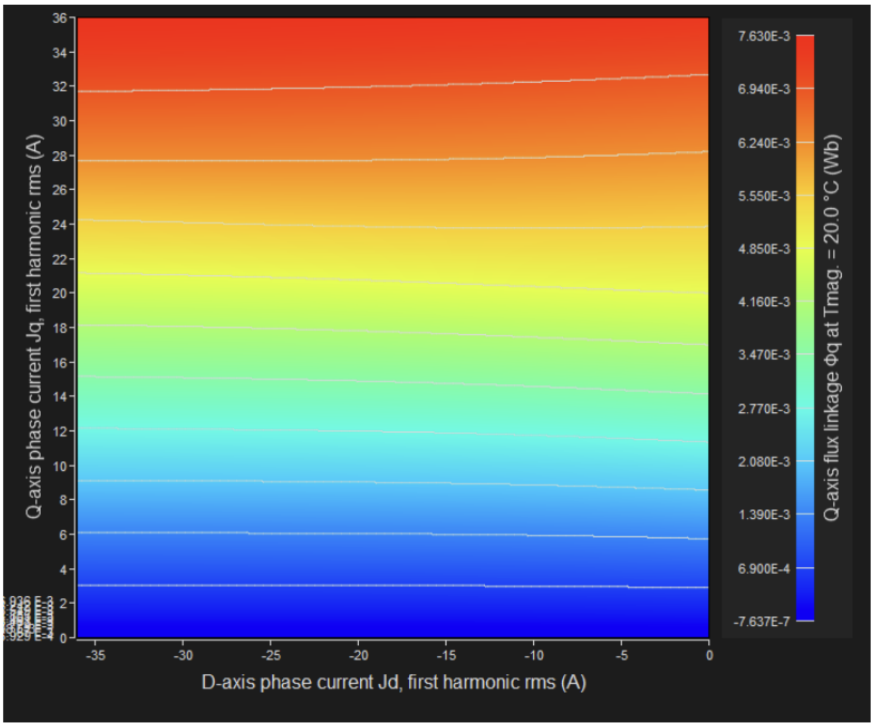

Figure 17. Q-axis flux linkage $\boldsymbol{\Phi}_{\boldsymbol{q}}$ in $\boldsymbol{J}_{\boldsymbol{d}}$-$\boldsymbol{J}_{\boldsymbol{q}}$ area

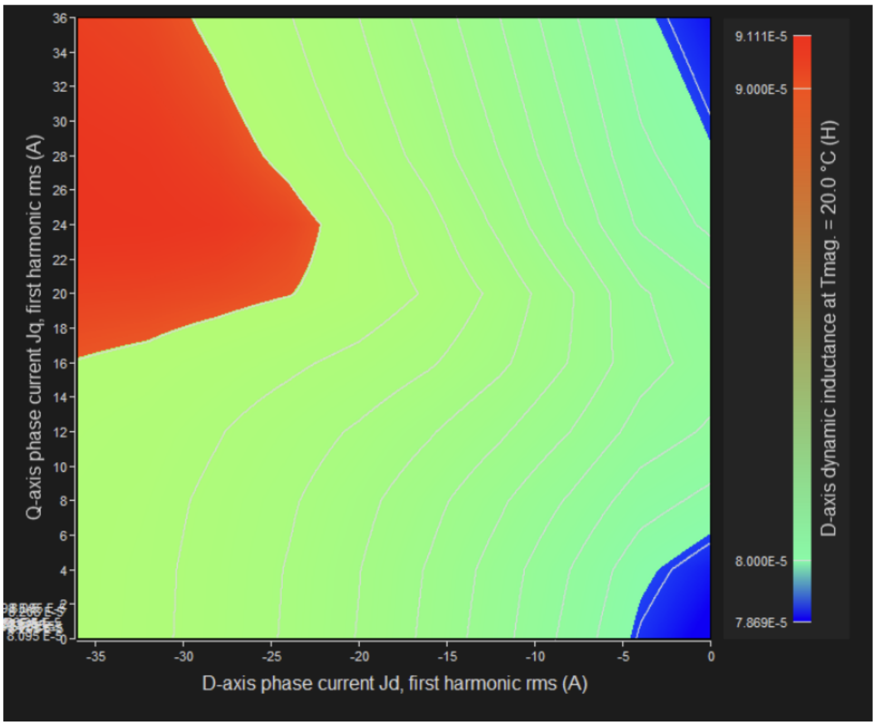

Figure 18. D-axis dynamic inductance L_d in J_d-J_q area

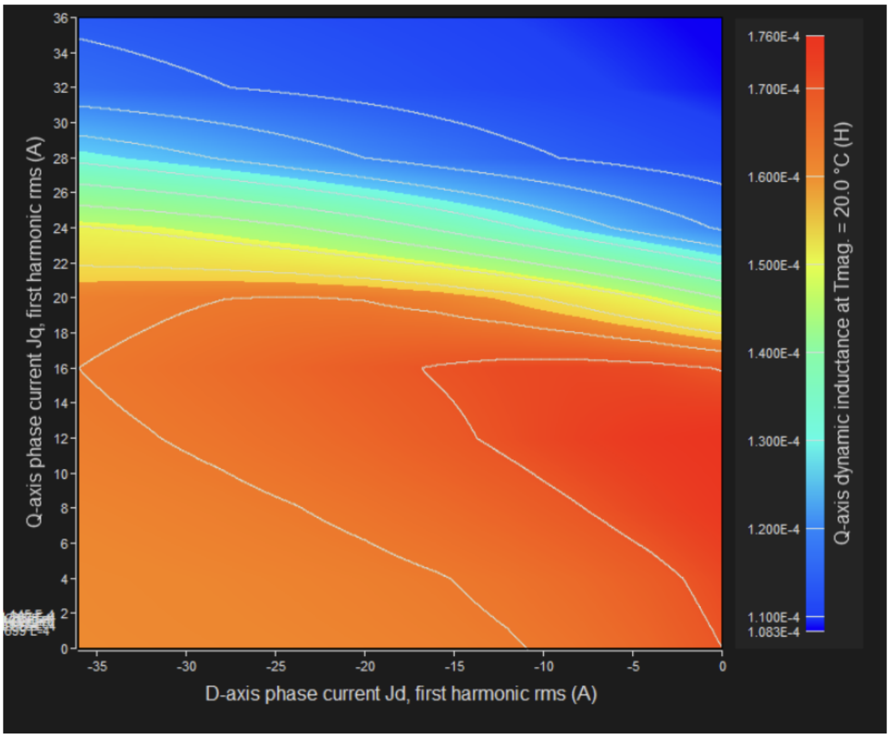

Figure 19. Q-axis dynamic inductance L_q in J_d-J_q area

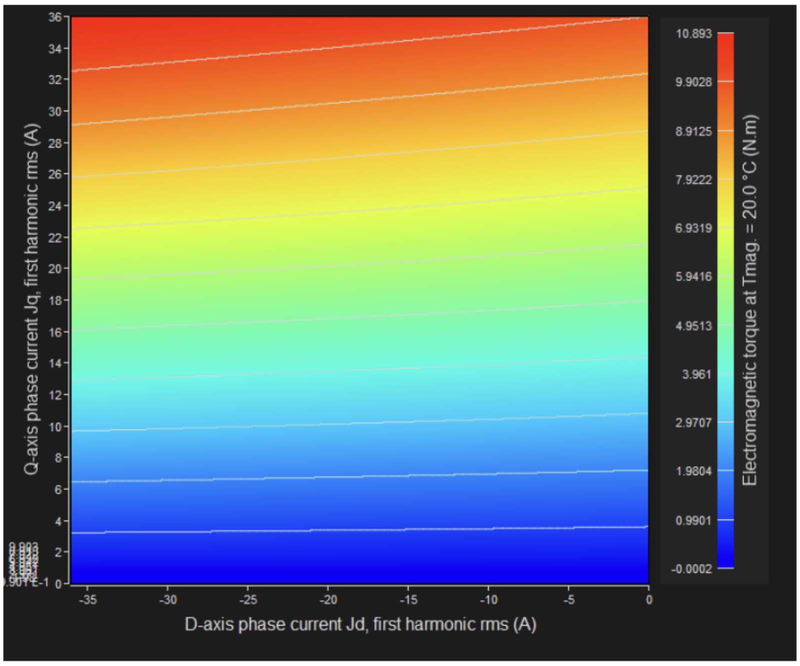

Figure 20. Electromagnetic torque in J_d-J_q area

The d-q axis parameter maps verify the IPM characteristics of the design and its non-linear behavior dependent on the current. In the next stage, the performance of this motor structure under a specific operating point was evaluated through a datasheet analysis.

7. Datasheet / Operating Point Analysis

A datasheet test was used to evaluate the summary performance of the motor under specific current and voltage limits. This test brings together information such as the motor's base speed point, fundamental electrical quantities, loss distribution, d-q model parameters, open-circuit equivalents, and mechanical torque ripple into a single operating point. Therefore, the datasheet analysis serves as an important summary test in terms of observing how the design behaves in a practical sense.

In this study, the datasheet analysis was performed under a maximum line current of 36 A RMS, a maximum line-to-line voltage of 390 V RMS, and the MTPV command mode. The winding and magnet temperatures were kept constant at 20 °C, and the mechanical loss model was left as zero by user definition. According to the obtained results, the base speed point of the motor was calculated to be approximately 23,042 rpm.

7.1 Base Speed Point and General Performance

According to the obtained operating point data, the motor produces a mechanical torque of approximately 8.989 N·m and provides 21.69 kW of mechanical power at this point. The machine's electrical power is 24.05 kW, and the total losses are at the level of 2.36 kW. Correspondingly, the efficiency was found to be approximately 90.17%. The power factor being at a level of 0.991 indicates that the motor uses active power effectively at this operating point.

The main point of interest here is that, despite the speed being quite high, the generated torque remains low. This situation indicates that the point in question is not the maximum torque point, but is related to the base speed / high-speed region where the voltage limit is reached. The fact that the open-circuit line-to-line voltage is obtained as approximately 385.4 V RMS at the same point, which is very close to the 390 V limit, confirms this interpretation. In other words, the motor is constrained by the voltage limit at this point, and the need for field weakening control emerges at higher speeds.

The fact that the control angle is only 5.195° and that, in the d-q model table, the Id component is small while the Iq component is dominant, shows that the torque production at this point is predominantly realized through the q-axis current. This situation suggests that the motor operates with a limited level of negative Id at this operating point and that the reluctance contribution, although present, is not dominant.

7.2 Loss Distribution

One of the most important findings of the datasheet analysis is the loss distribution. According to the results, the total loss is 2,363 W, of which approximately 2,284 W consists of iron losses, and only 79.7 W consists of stator Joule losses. Mechanical losses, on the other hand, were defined as zero in the analysis. This distribution clearly demonstrates that iron losses become dominant in the high-speed region.

The fundamental reason for the iron losses being this high is that the electrical frequency reaches a level of 1,536 Hz. at the base speed point. In other words, the main cause of the decrease in efficiency is not copper losses, but electrical steel losses increasing under high frequency. This finding indicates that the design's performance in the high-speed region must be evaluated not only in terms of torque but also in terms of losses and heating.

7.3 Flux Density and Magnetic State

When the magnetic results under load are examined, the peak value of the air gap flux density was found to be 1.096 T. The flux density occurred in the range of approximately 1.69–1.76 T in the stator tooth and yoke regions, and at the level of 1.74–1.79 T in the rotor web and rotor yoke regions. In particular, obtaining values of 2.081 T and 2.249 T in the rotor bridge and rotor pole shoe regions, respectively, indicates the presence of significant local saturation in these regions.

Compared to the open-circuit condition, it is seen that the flux density in the rotor web region increases significantly under load. This situation demonstrates that the stator current strongly influences the internal magnetic paths of the rotor. This concentration, occurring especially in narrow cross-section rotor regions, can also be impactful on torque production, harmonic content, and losses.

From the perspective of magnet behavior, it is seen in the electromagnetic analysis that the magnets do not carry a risk of demagnetization. The results obtained according to the 50% and 90% coercive field limits show that the magnets remain in the safe zone at this operating point. However, this interpretation is valid only for electromagnetic analysis conditions; considering the high magnet temperatures obtained in the thermal analysis, this situation must be additionally evaluated under actual operating conditions.

7.4 Inductances and d-q Model Representation

In the datasheet results, both unsaturated inductances and d-q inductances belonging to the operating point are given. In the unsaturated condition, it is seen that the L_d and L_q values are quite close to each other. In contrast, at the base speed point, the d-axis inductance $L_d = 7.984 \times 10^{-5}$ H, and the q-axis inductance $L_q = 1.084 \times 10^{-4}$ H were obtained. This difference shows that the $L_q > L_d$ relationship becomes more pronounced under the operating point.

This result is consistent with the IPM character observed in the previous map analyses. In other words, rotor saliency demonstrates its effect not only theoretically but also under the operating point. However, at this point, it is seen that the I_d component is quite limited and a large part of the torque production is determined by I_q.

7.5 Torque Ripple

One of the most striking results obtained in the datasheet test is the mechanical torque ripple. While the average mechanical torque is 8.989 N·m, the peak-to-peak value of the ripple mechanical torque was found to be 4.648 N·m. This value corresponds to approximately 51.7% of the average torque. The ratio in question is quite high and can create a negative situation in terms of vibration, acoustic behavior, and driving comfort, especially in traction applications.

There could be multiple reasons for this high ripple level. The cogging torque observed in the previous open-circuit analyses, harmonic interactions occurring between the rotor and stator, the lack of skew application, and the deviation of the air gap flux density from the ideal sine form can be considered as the main factors contributing to this result. Furthermore, since the average torque level is relatively low at this operating point, the absolute ripple amplitude appears larger as a percentage. Despite this, a peak-to-peak ripple of 4.648 N·m is at a level that must be taken into consideration as an absolute magnitude.

7.6 General Evaluation

As a result of the datasheet test, it was seen that the motor can operate in the high-speed region, can provide approximately 90% efficiency at a point close to the voltage limit, and has a high power factor. From this perspective, it can be said that the motor does not exhibit a completely inconsistent structure electromagnetically. However, the same test also clearly reveals some weak aspects of the design.

Firstly, the fact that a very large portion of the losses consists of iron losses indicates that the high-speed region is critical for the design. Secondly, the high torque ripple indicates that the design needs to be improved in terms of harmonic and mechanical smoothness. Thirdly, the local saturation observed in the rotor bridge and pole shoe regions shows that magnetic cross-sections are strained in certain regions. Therefore, the operating point analysis demonstrates that the motor is fundamentally workable; however, it requires further optimization in terms of electromagnetic quality, loss level, and torque smoothness.

Table 10. Fundamental performance results obtained in the datasheet analysis.

Parameter	Value
Operating mode	Motor
Mechanical torque	8.989 N·m
Speed	23 041.977 rpm
Mechanical power	21 689.431 W
Electrical power	24 052.892 W
Total loss	2 363.461 W
Efficiency	%90.174
Power factor	0.9909
Line current	36.0 A RMS
Line-to-line voltage	390.0 V RMS
Electrical frequency	1 536.132 Hz
Control angle	5.195°

Table 11. Loss distribution in the datasheet analysis

Loss component	Value
Total loss	2 363.461 W
Stator Joule loss	79.686 W
Iron loss	2 283.775 W
Mechanical loss	0.0 W

Table 12. Magnetic state and ripple results in the datasheet analysis

Parameter	Value
Air gap flux density, ARV	5.383×10^{-1} T
Air gap flux density, peak	1.096 T
Stator tooth, max	1.755 T
Stator yoke, max	1.688 T
Rotor yoke, max	1.738 T
Rotor web, max	1.792 T
Rotor bridge, max	2.081 T
Rotor pole shoe, max	2.249 T
Ripple torque, peak-to-peak	4.648 N·m
Ripple torque / average	%51.705
Ripple torque period	15°

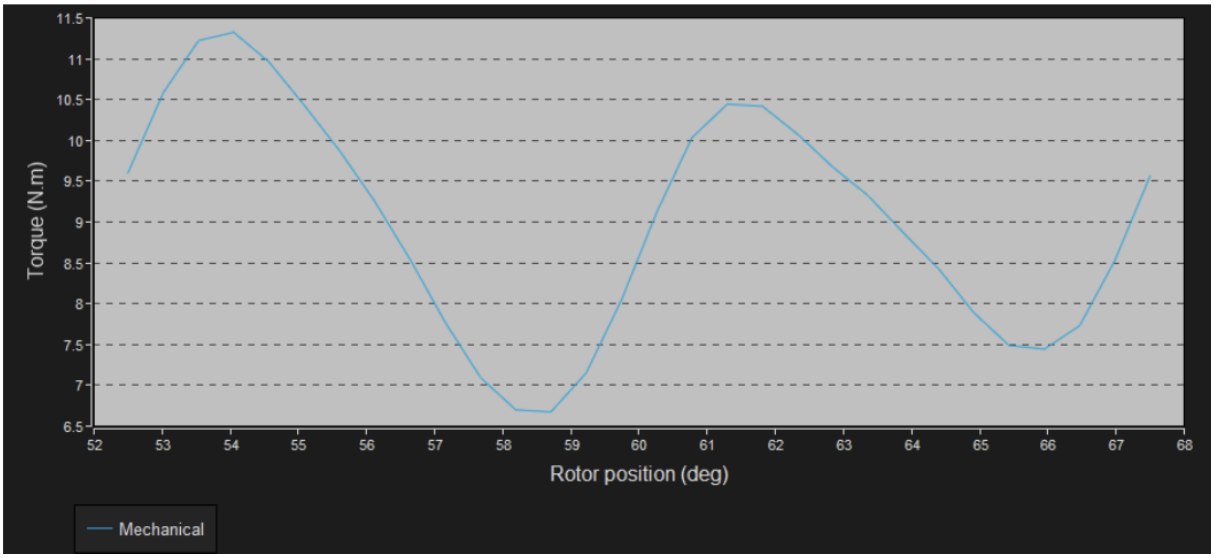

Figure 21. Ripple mechanical torque versus rotor angular position

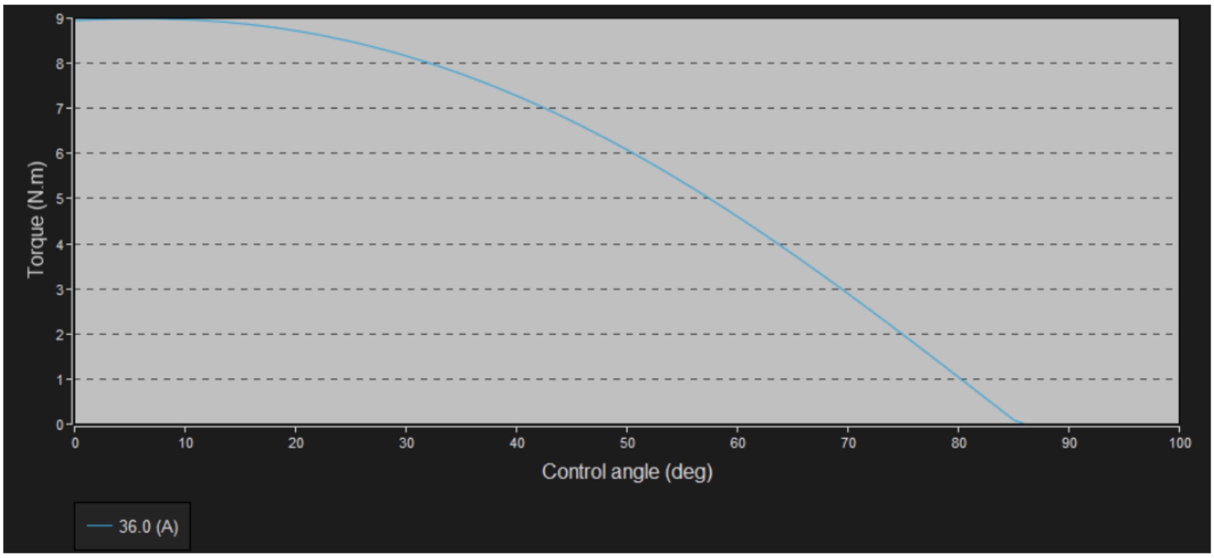

Figure 22. Mechanical torque versus current and control angle

The datasheet analysis revealed the general performance of the motor in the high-speed region; while yielding acceptable results in terms of efficiency and power factor, it indicated that the design is open to improvement in terms of iron loss and torque ripple. This situation necessitated a separate investigation of thermal performance in particular.

8. Thermal Analysis and Thermal Evaluation of the Design

The mere electromagnetic operability of a motor design is not sufficient for traction applications. Especially in electric vehicle motors, winding temperature, magnet temperature, heat dissipation through the housing, and temperature distribution in the inner regions are decisive for the sustainability of the design. Therefore, a steady-state thermal analysis was performed following the electromagnetic analyses.

In this study, the steady-state thermal analysis was conducted under the conditions of a 750 rpm speed, 770 W stator Joule loss, and 240 W stator iron loss. Rotor iron losses, magnet losses, and mechanical losses were assumed to be zero. The ambient temperature was taken as 20 °C; the cooling structure was limited to natural convection, radiation, and the internal air conduction model, as defined previously. Therefore, the obtained temperature distributions demonstrate how the selected passive cooling approach behaves under these losses.

8.1 General Overview of Temperature Distributions

When the radial and axial temperature distributions are examined, it is clearly seen that the highest temperature regions are the stator windings and particularly the winding ends. In the radial view, the conductor regions within the slot reach hotter colors, while in the axial view, it is understood that heat accumulation occurs in the end winding regions on both the connection and opposite connection sides. This distribution confirms that the primary heat source within the motor is the Joule losses generated in the copper regions.

In contrast, the housing, end-cap, and regions close to the outer surface remain at lower temperature levels. This situation indicates that heat can be transferred from the inside to the outside, but the heat generation in the inner regions remains excessive compared to the existing thermal path and cooling capacity. Consequently, the problem emerges not only on the outer surface but also in the heat transfer paths from the winding to the housing and from the end winding regions to the environment.

8.2 Critical Temperature Regions

Upon examining the final temperature results, it is observed that the highest temperatures occur in the end winding regions. The connection side end winding temperature was calculated as 387.239 °C, and the opposite connection side end winding temperature as 387.774 °C. The in-slot winding temperature reached a level of 375.634 °C. The stator tooth temperature was found to be 307.594 °C, and the tooth foot temperature was 311.323 °C. These values indicate that there are significant limitations in transferring the heat from the winding regions to the magnetic circuit.

On the rotor side, the magnet temperature was obtained as approximately 289.006 °C, the rotor web temperature as 289.922 °C, the bridge temperature as 291.999 °C, and the pole shoe temperature as 290.976 °C. These results show that the rotor also does not remain at low temperatures; the heat originating from the stator is also transferred to the rotor side across the air gap. The rise of the air gap temperature to a level of 304.842 °C supports this interpretation.

The housing temperature was calculated to be approximately 268,61 °C, and the shaft temperature was 280.367 °C. Despite the ambient temperature being 20 °C, the housing remaining at such a high temperature indicates that the amount of heat transferred to the external environment via natural convection is insufficient.

8.3 Interpretation of Thermal Results

These results clearly demonstrate that the design, in its current state, does not possess a thermally robust structure. The most critical point is the concentration of temperature in the copper regions and winding ends. The main reasons for this include the absence of cooling fins on the housing, the lack of an active cooling circuit, the limitation of external and internal heat transfer solely to natural convection, the absence of impregnation within the slot, and the resulting low equivalent radial thermal conductivity of the slot.

In particular, the low equivalent radial thermal conductivity in the slot model prevents the heat generated in the winding from being effectively transferred to the stator teeth and the housing. Therefore, the temperature accumulates in the in-slot conductor and end winding regions. The fact that the end winding regions are the hottest points is an expected result in this regard; because Joule losses occur in this region, yet the heat dissipation remains weaker as the thermal path is not directly connected to the housing.

The magnet temperature reaching a level of approximately 289 °C constitutes a separate problem. Although the magnets appear to be in a safe zone in the electromagnetic analysis, according to the material data of the NdFeB magnet used, the maximum operating temperature is 120 °C. Therefore, the magnet temperature reached in the thermal analysis is high enough to pose a risk of irreversible property loss and demagnetization in a real operating scenario. For this reason, when the electromagnetic and thermal results are evaluated together, it is understood that magnet safety is restricted not only in terms of the magnetic field but also in terms of temperature.

8.4 Impact of the Cooling Structure on the Results

The cooling approach selected for the design has a significant impact on the high thermal results obtained. Leaving the fin parameters as “None” in the design report limits the external surface area of the housing. Similarly, selecting the external cooling mode as natural convection and not defining an active cooling circuit leads to a low amount of heat that can be dissipated from the housing to the environment. Consequently, even the housing temperature reaches a considerably high level.

On the internal cooling side, the use of only natural convection and radiation mechanisms makes it difficult for the temperatures generated in the end-space regions to decrease. The low equivalent radial conductivity within the slot and the absence of impregnation further increase the winding temperatures. Therefore, the high temperatures are a result of not only the loss levels but also the selected passive thermal design approach.

8.5 Thermal Evaluation of the Design

Thermal analysis results indicate that although the first design iteration is electromagnetically functional, it is not thermally sustainable. In particular, the winding and magnet temperatures have significantly exceeded typical motor insulation classes and the operating limits of NdFeB magnets. Therefore, it cannot be stated that this design in its current form is suitable for continuous operation.

However, this result does not imply that the design is entirely meaningless. On the contrary, it clearly reveals which regions are thermally critical. The end winding, in-slot winding, and rotor magnet regions are areas that require direct improvement in the next design iteration. It is assessed that thermal performance can be significantly enhanced if cooling fins are added to the housing, an active cooling circuit is utilized, impregnation/potting is applied, the end winding thermal path is improved, and electromagnetic optimizations are performed to reduce high-speed losses.

Table 13. Steady-state thermal analysis inputs

Parameter	Value
Speed	750 rpm
Stator Joule loss	770 W
Stator iron loss	240 W
Magnet loss	0 W
Rotor iron loss	0 W
Mechanical loss	0 W
Ambient temperature	20 °C

Table 14. Critical temperature regions in steady-state thermal analysis

Region	Temperature
C.S. end winding	387.239 °C
O.C.S. end winding	387.774 °C
In-slot winding	375.634 °C
Tooth	307.594 °C
Tooth foot	311.323 °C
Yoke (stator)	292,66 °C
Magnet	289.006 °C
Rotor web	289.922 °C
Rotor bridge	291.999 °C

Region	Temperature
Rotor pole shoe	290.976 °C
Frame	268,61 °C
Airgap	304.842 °C

Table 15. Brief evaluation of thermal results

Analyzed Element	Observation	Interpretation
End winding temperature	Highest temperature region	Weak heat dissipation
In-slot winding temperature	Very high	Low radial thermal conduction
Magnet temperature	Critically high	Risky for actual operation
Housing temperature	High	Insufficient external cooling
Airgap temperature	High	Strong stator-rotor thermal interaction
General result	Thermally weak first iteration	Cooling improvement required

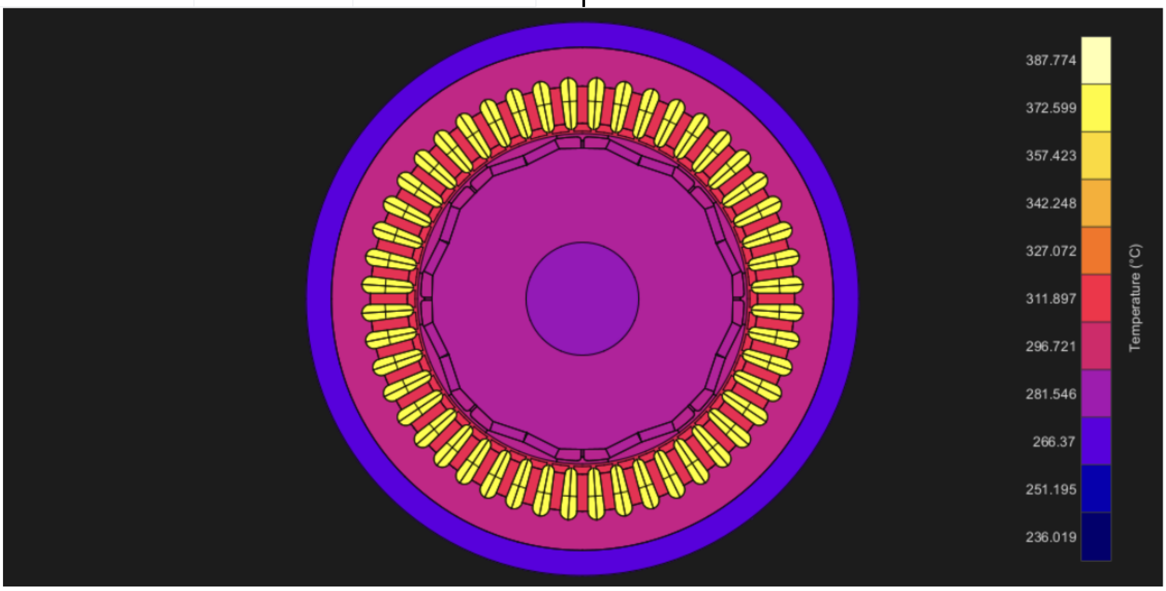

Figure 23. Temperature radial view

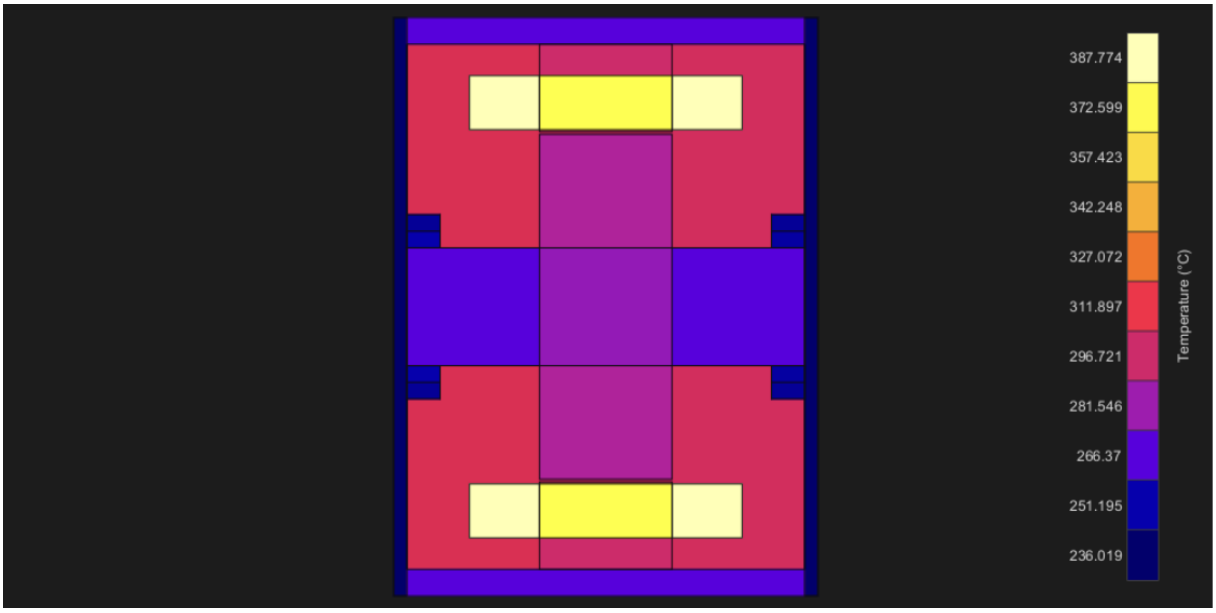

Figure 24. Temperature axial view

The thermal analysis results showed that the most critical problem in the first iteration of the design is the winding and magnet temperatures. For this reason, in the next section, the driver-control structure associated with the electromagnetic model was examined, and a PSIM-based evaluation was conducted regarding the dynamic operation of the motor.

9. IPM Motor Drive Model in PSIM Environment

In order to investigate the drive and control side behavior of the Nissan Leaf referenced IPM motor structure created in the FluxMotor environment, a motor drive model was established in the PSIM environment. In this model, a three-phase two-level voltage source inverter, a speed control loop, a d-q axis current control loop, and reference current generation blocks dependent on the operating region are used together. Thus, not only the static electromagnetic character of the motor but also its dynamic operating behavior under a drive has become observable.

The established system is supplied with a 700 V DC bus. A 10 kHz switching frequency is used on the inverter side. The motor model was driven in a closed loop with three-phase current measurements and mechanical speed feedback. In the system, the load torque is defined as 150 N·m, and the load moment of inertia as 0.01. Motor parameters such as phase resistance, d-q axis inductances, number of poles, moment of inertia, and speed limit are defined in the model inputs.

The drive structure consists of two main control loops. The speed controller located in the outer loop generates a current command based on the error between the reference speed and the measured speed. In the inner loop, I_d and I_q currents are regulated using d-q axis current controllers. The three-phase currents are transformed to the d-q axis with the help of transformation blocks, the reference and measured currents are compared to obtain voltage commands, and these commands are applied to the inverter via the PWM block.

The most important part of the PSIM model in this study is the control approach used in the reference current generation block. A control block comprising maximum torque per ampere (MTPA), field weakening, and maximum torque per voltage (MTPV) strategies is used in the motor control structure. Thanks to this structure, the motor can be operated in the MTPA region where the current is used more efficiently at low speeds, and in the field weakening region at higher speeds to remain under the voltage limit. The MTPV approach is also included in the control structure for operating conditions close to the high speed and voltage limits.

In the simulations, the reference speed was selected as 8000 rpm. When the obtained speed graph is examined, it is seen that the motor speed rises smoothly from the beginning and reaches the reference value in approximately 0.135 s. After reaching the reference value, it is understood that the speed is kept stable at this level. This result shows that the outer speed loop can track the desired operating point.

When the d-q axis current responses are examined, it is observed that the I_q component is at a higher level in the initial stage and gradually decreases as the speed increases. In contrast, it is observed that the I_d current shifts to more negative values. This behavior indicates that the q-axis current is dominant for torque production in the low-speed region; and as the speed increases, field weakening is performed by applying a negative I_d due to the effect of the inverter voltage limit. The fact that the reference currents and the actual currents are generally close to each other indicates that the current control loop operates at an acceptable level.

PSIM simulation results present a control behavior consistent with the IPM character obtained in the FluxMotor environment. It is also seen in this study that an MTPA and field weakening based driving structure is suitable for a motor with an $L_q > L_d$ relationship in the d-q axis. In particular, shifting the I_d component in a more negative direction as the speed increases is an expected result for the motor to be driven under voltage limits in the high-speed region.

Consequently, the established PSIM model provides a suitable drive infrastructure for the selected Nissan Leaf referenced IPM motor topology. The ability to track the speed reference, the d-q currents generally following their references, and the observation of the field weakening behavior as the speed increases demonstrate that the driver-control structure is functional. Since the aim within the scope of this study is to establish a basic drive structure compatible with the motor model and to observe the behavior of the control strategies, rather than a detailed control optimization, the obtained results are considered acceptable for this purpose.

Table 16. PSIM drive model main parameters

Parameter	Value
DC bus voltage	700 V
Switching frequency	10 kHz
Maximum inverter current	400 A
Load torque	150 N·m
Load moment of inertia	0.01
Phase resistance, R_s	0.02035 Ω
d-axis inductance, L_d	0.0001711 H
q-axis inductance, L_q	0.0004245 H
Number of poles	8
Moment of inertia, J	0.0206
Shaft torque parameter	100
Maximum speed	10000 rpm
Reference speed	8000 rpm
Current sampling frequency	10 kHz
Speed sampling frequency	10 kHz

Table 17. Key observations from PSIM simulation

Analyzed element	Observation	Interpretation
Speed response	8000 rpm reference is reached	Speed loop is working
I_q behavior	High initially, decreases with speed	Torque production is dominant at low speed
I_d behavior	Becomes more negative as speed increases	Field weakening effect
Current reference tracking	I_d and I_q are generally consistent with I_{dr} and I_{qr}	Inner current loop is functional
High-speed region	Stable operation around reference speed	Drive structure is appropriate

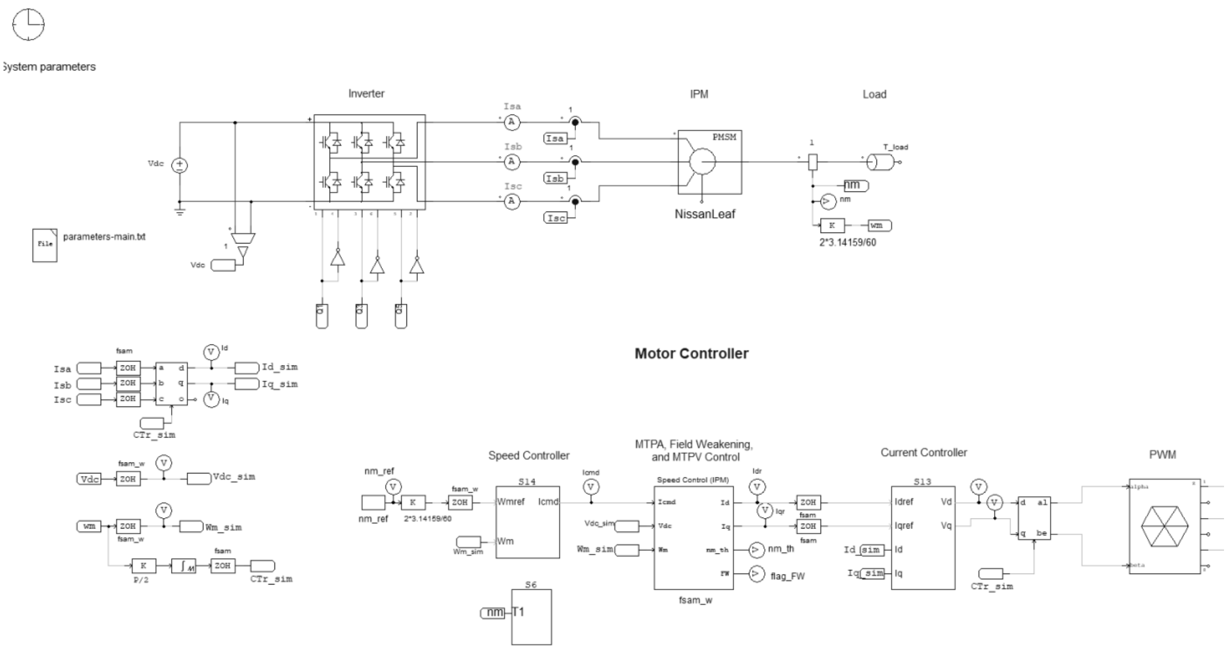

Figure 25. General block diagram of PSIM IPM drive model

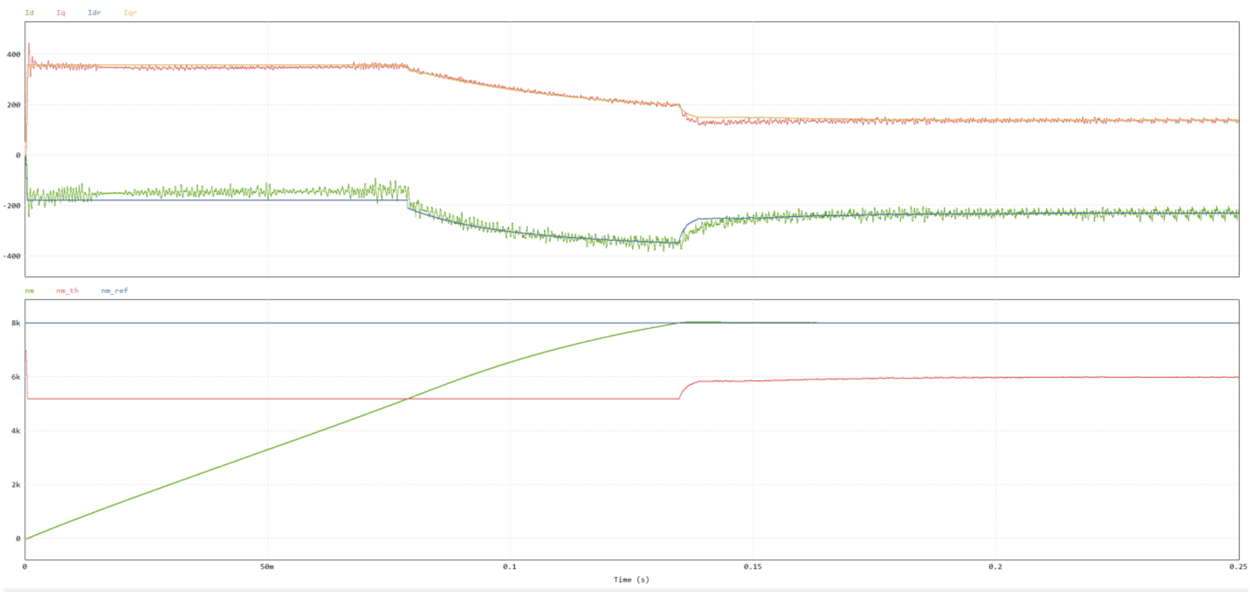

Figure 26. Speed response and d-q current responses

9.1 Simulation Results

In the simulation performed in the PSIM environment, an 8000 rpm reference speed was applied to the motor. According to the obtained speed graph, the motor speed increased smoothly from the beginning and reached the reference value in approximately 0.135 s. After reaching the reference value, it is observed that the speed is maintained at this level. This result indicates that the speed control loop is operating and the system can track the reference speed.

When the d-q axis current responses are examined, it is observed that the I_q component is high in the initial region and decreases as the speed increases. In contrast, the I_d current shifts to more negative values. This behavior corresponds to the MTPA character in the low-speed region and the field weakening behavior in the high-speed region. The fact that the reference currents and the measured currents are generally close to each other indicates that the current control loop operates at an acceptable level.

In general, the PSIM simulation demonstrated that the established drive structure, together with the Nissan Leaf referenced IPM motor model, can provide basic speed and current control.

10. General Evaluation

In this study, a Nissan Leaf referenced IPM motor topology was created in the FluxMotor environment, its electromagnetic character was examined through open-circuit and operating point analyses, and subsequently, a driver-control model was established in the PSIM environment for the same motor type. The obtained results show that the selected topology exhibits physically consistent IPM behavior. Back-EMF results, d-q axis parameter maps, and the $L_q > L_d$ relationship confirm that the motor possesses a saliency effect and a reluctance torque contribution.

However, in the operating point analysis, it was observed that iron losses became dominant in the high-speed region, and the mechanical torque ripple was high. This situation indicates that although the design is electromagnetically functional, it is open to improvement in terms of harmonic content and torque smoothness. In the thermal analysis, high temperature values were obtained, particularly in the end winding, in-slot winding, and magnet regions. When evaluated along with the finless housing structure, the absence of an active cooling circuit, the natural convection assumption, and the non-impregnated winding structure, this result demonstrates that the design is thermally weak in its current state.

The driver model established on the PSIM side showed that the speed reference was tracked and the d-q axis current control fundamentally worked. The introduction of the negative I_d component in the current responses as the speed increases reveals that field weakening behavior is observed. In this respect, the control structure yielded a result consistent with the motor character obtained in the FluxMotor analyses.

Overall, the study enabled the joint evaluation of the first design iteration of a Nissan Leaf type IPM motor structure in terms of electromagnetics, thermals, and control. The obtained results clearly revealed the aspects of the design that need development as well as its strong points.

11. Conclusion

In this study, a motor model with an interior permanent magnet synchronous motor structure referenced to the Nissan Leaf was created in the FluxMotor environment, and its basic electromagnetic and thermal analyses were performed. The obtained results showed that the motor exhibited an IPM motor character in terms of open-circuit behavior, d-q axis parameters, and operating point performance. In addition, speed control, d-q current control, the MTPA approach, and field weakening behavior were investigated with the driver model established in the PSIM environment.

As a result of the analyses, it was observed that the motor offered an electromagnetically operable structure, but it was open to improvement, especially in terms of torque ripple, iron losses in the high-speed region, and thermal performance. The concentration of high temperatures particularly in the winding and winding end regions during the thermal analysis indicated that the selected passive cooling structure was inadequate. In contrast, the PSIM simulation results revealed that the established driver structure could track the reference speed and exhibit control behavior compatible with the motor character.

Consequently, this study presented a practical evaluation that simultaneously examined the design, analysis, and driver control process of a Nissan Leaf type IPM motor structure. In future studies, a more balanced motor performance can be achieved by improving the cooling structure, reducing harmonics and torque ripple, and optimizing the electromagnetic design.

This study demonstrated the necessity of handling a Nissan Leaf referenced IPM motor structure not only from an electromagnetic perspective but also collectively in terms of thermals and driver control.

12. References

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